

①

$$A = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}$$

09.20

a) EW exakt

b) EW mit Gerschgorin

c) Vektoriteration: $x^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ bis $\mu^{(2)}$

$$\begin{aligned} a) \quad 0 &= \det(A - \lambda I) = \det \begin{pmatrix} 3-\lambda & 4 \\ 4 & 3-\lambda \end{pmatrix} \\ &= (\lambda-3)^2 - 4^2 \end{aligned}$$

$$(\lambda-3)^2 = 4^2$$

$$\lambda_{1,2} - 3 = \pm 4$$

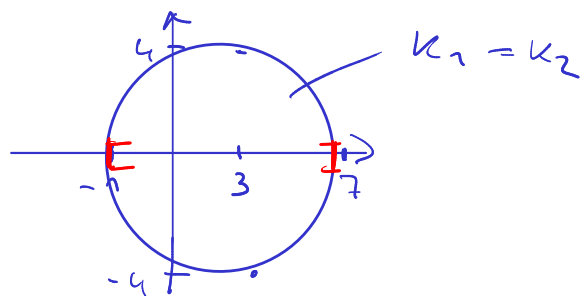
$$\lambda_{1,2} = 3 \pm 4$$

$$\lambda_1 = \underline{\underline{7}}, \quad \lambda_2 = \underline{\underline{-1}}$$

$$b) \quad \underline{\text{Gersch.}}: \quad A = \begin{pmatrix} \underline{3} & \underline{4} \\ \underline{4} & \underline{3} \end{pmatrix} \leftarrow m_1 = 3, \quad r_1 = 4$$

$$\leftarrow m_2 = 3, \quad r_2 = 4$$

$$K_1 = K_2 = \{ z \mid |z-3| \leq 4 \}$$



$$I = [-1, 7]$$

c) Varianz: $x^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $A = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}$

$$y^{(k+1)} = A x^{(k)}$$

$$x^{(k+1)} = \frac{y^{(k+1)}}{\|y^{(k+1)}\|_2}$$

$$\mu^{(k)} = (y^{(k+1)}, x^{(k)})_2$$

iter. 1: $y^{(1)} = A x^{(0)} = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$$x^{(1)} = \frac{y^{(1)}}{\|y^{(1)}\|_2} = \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$\mu^{(1)} = (y^{(1)}, x^{(0)}) = \begin{pmatrix} 3 \\ 4 \end{pmatrix}^T \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \underline{\underline{3}}$$

iter. 2: $y^{(2)} = A x^{(1)} = \begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix} \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

$$= \frac{1}{5} \begin{pmatrix} 25 \\ 24 \end{pmatrix}$$

$$x^{(2)} = \frac{y^{(2)}}{\|y^{(2)}\|_2} = \frac{1}{c} \begin{pmatrix} 25 \\ 24 \end{pmatrix}, \quad \frac{1}{c} = \sqrt{25^2 + 24^2}$$

$$\mu^{(2)} = (y^{(2)}, x^{(1)})_2$$

$$= \frac{1}{5} \begin{pmatrix} 25 \\ 24 \end{pmatrix}^T \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

$$= \frac{25 \cdot 3 + 24 \cdot 4}{25} = \frac{75 + 96}{25}$$

$$= \frac{171}{25}$$

$$= \frac{684}{100} = \underline{\underline{6.84}}$$

$$\left| \frac{\lambda_2}{\lambda_1} \right| = \frac{1}{7} \quad O\left(\left(\frac{1}{7}\right)^{2k}\right)$$